

PRESEASON VARIATIONS IN AEROBIC FITNESS AND PERFORMANCE IN ELITE-STANDARD SOCCER PLAYERS: A TEAM STUDY

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ABSTRACT

Castagna, C, Impellizzeri, FM, Chaouachi, A, and Manzi, V. Preseason variations in aerobic fitness and performance in elite-standard soccer players: A team study. *J Strength Cond Res* 27(11): 2959–2965, 2013—The aim of this study was to examine the effects of individual training loads considered as permanent in selected heart-rate (HR) zones on aerobic fitness and performance in elite professional soccer players. Eighteen professional soccer players were observed during the prechampionship training period (8 weeks). Speeds and HR at 2 and 4 mmol·L⁻¹ blood-lactate concentrations (S2, S4, respectively), $\dot{V}O_{2\max}$, and Yo-Yo intermittent recovery test level 1 performance (Yo-Yo IR1) were assessed pretraining and posttraining. Training intensities were categorized using 3 HR zones: low intensity (<HR 2 mmol·L⁻¹), moderate intensity (between HR 2 and 4 mmol·L⁻¹), and high intensity (>HR 4 mmol·L⁻¹). Training-session HRs ($n = 900$) showed a polarized distribution with 73.6 ± 3.7 (2,945 $\pm$ 148 minutes), 19.1 ± 3.5 (763 $\pm$ 141 minutes), and $7.3 \pm 2.9\%$ (292 $\pm$ 116 minutes) of the total training time spent at low, moderate, and high intensities, respectively ($p < 0.001$). The S2 and S4 significantly improved posttraining (+10 and 7%, respectively, $p < 0.001$). The $\dot{V}O_{2\max}$ and Yo-Yo IR1 values were 6 and 19.5% higher posttraining, respectively ($p < 0.01$). Training performed at high intensity was significantly related to relative improvement in S2 ($r = 0.78$, $p = 0.002$), S4 ($r = 0.60$, $p = 0.03$), $\dot{V}O_{2\max}$ ($r = 0.65$, $p = 0.02$), and Yo-Yo IR1 ($r = 0.66$, $p = 0.01$). The results of this study provided further evidence for HR longitudinal validity and effectiveness of the high-intensity training (i.e., >90% HRmax) in men's professional soccer. In this regard, the time

spent at high intensity should be in the range of 7–8% of the total training time during preseason.

KEY WORDS association football, training load, lactate threshold, team sports, heart rate

INTRODUCTION

Elite-standard soccer requires a well-developed aerobic fitness level to be successfully played (31). Indeed game demands locate in 70–80% of maximal oxygen uptake ($\dot{V}O_{2\max}$) with an average exercise heart rate (HR) that was similar to that reported for usual players' anaerobic threshold (i.e., 80–90% HRmax) (31).

Training studies showed that improvement in aerobic fitness in the range of 8–12% was effective in promoting match activities without impairment of neuromuscular performance (20,31). As a general rule, a $\dot{V}O_{2\max}$ of 60 ml·kg⁻¹·min⁻¹ was suggested as the minimum fitness requirement to successfully play at the men's professional level (31).

Because of the limited access to professional teams, experimental (i.e., randomized controlled trial) aerobic-training intervention studies reporting short-term effects (4–8 weeks) on match performance were only possible with young soccer players (17,18,20). Given this, Castagna et al. (4) proposed longitudinal-descriptive training studies (i.e., no experimental intervention provided). This was done to gain information that is helpful for aerobic-fitness development in Elite-standard soccer players. In this study, Castagna et al. (4) showed that interindividual responses to team training accounted for the reported individual improvement in aerobic fitness assumed as speed at selected blood-lactate concentrations (i.e., 2 and 4 mmol·L⁻¹). This was possible because differently from endurance sports, where training prescription is mainly individual, soccer players usually train in a group (7,20,27). Consequently, players' physiological responses to training load (TL) may present remarkable interindividual variations (24,30).

Match-to-match variation in game activities has been suggested as a limitation for assessing the effect of fitness improvement on match activities in soccer training studies

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(16). To limit across-match variability, the use of a criterion-performance test may be of help in tracking a player's ability to cope with game demands (1). In this regard, the Yo-Yo intermittent recovery test in its level 1 version (i.e., Yo-Yo IR1) was reported as a relevant measure of mainly aerobic intermittent high-intensity endurance in soccer (i.e., criterion-convergent validity) (1). Furthermore, the Yo-Yo IR1 was shown to be related to match activities performed at high intensity assumed as a key variable in competitive soccer (i.e., construct convergent validity) (1,3,5). Given this, the Yo-Yo IR1 may be used to estimate players' capability to perform at a high intensity during the game.

The assessment of the individually experienced TL (i.e., internal load) is vital in professional soccer as per mainly team nature of the provided training stimuli (5). Despite the number of successful methods developed to track the internal load of players, HR monitoring still provides a number of practical advantages (4,12,21,28).

Indeed HR monitoring has proved to be valid in tracking acute variation in oxygen uptake during soccer drills (12). Additionally, permanence in selected HR zones was reported to account for improvements in submaximal aerobic-fitness variables providing longitudinal validity for HR monitoring in Elite-standard professional soccer (4).

Unfortunately, no information was reported in the international literature as per the longitudinal validity of HR monitoring in tracking changes in maximal variables of aerobic fitness (i.e., $\dot{V}O_{2\max}$) and performance (i.e., Yo-Yo IR1) in Elite-standard soccer players. Information in this regard could be of practical interest for TL analysts in soccer (4).

Therefore, the aim of this study was to examine the effect of permanence in selected HR zones on maximal and submaximal aerobic-fitness variables and match criterion performance (i.e., longitudinal validity) in Elite-standard soccer players during the preseason training period. The existence of an association between training intensity (i.e., time spent in selected HR zones) and reported changes in aerobic fitness and performance was assumed as the work hypothesis (4). The possible information provided by this study's results may potentially have a considerable practical impact on scientific coaching.

METHODS

Experimental Approach to the Problem

Despite the interest of the ecological and naturalist approach proposed by Castagna et al. (4), the convenience sampling (i.e., team and club study) and the small size of the elective cohorts addressed limit the external validity of this design and consequently the study result generalization. A practical strategy to solve this design bias maintaining group focus (i.e., elite level) in an ecological setup is inductive evidence. This was done with study replication under environmental (ecological setup) and experimental (i.e., similar independent variables) analogous conditions. Given this, this study was carried out with the aim to pro-

vide further evidence on the practical interest of HR zone training in soccer.

This descriptive nonexperimentally controlled training study was carried out during the precompetitive season of a men's Elite-Standard professional soccer team (i.e., Italian Serie A). In this period (8 weeks) of the season, the training routine mainly addressed the physical fitness development of players (4).

According to Castagna et al. (4), the TL was assessed by monitoring exercise HRs experienced by the outfield players over every single training session ($n = 900$) performed during the preparation phase of the competitive season (13,28). Because of the observational nature of the study, no training intervention or advice was provided by the authors of this research to the team staff.

Training intensity was quantified using 3 HR intensity zones based on the result of submaximal treadmill testing (4,28).

As physiological anchors were considered the HR attained at selected blood-lactate concentrations (i.e., 2 and 4 mmol·L⁻¹) (4,28). Training HR was categorized as low intensity (<HR at 2 mmol·L⁻¹), medium intensity (between HR at 2 and 4 mmol·L⁻¹), and high intensity (>HR at 4 mmol·L⁻¹) (4).

In this study, aerobic-fitness variables were considered to be $\dot{V}O_{2\max}$ and the running speed at 2 and 4 mmol·L⁻¹ (i.e., S2 and S4, respectively). These variables were considered as previously reported for their relevance with match performance and training responsiveness in men's soccer (4,25,31).

The capability of the observed players to cope with game demands was estimated using the Yo-Yo IR1 as performance criterion (1). This was done with the aim to estimate possible players' variations in match high-intensity activity as a consequence of interindividual TL variability.

Subjects

Eighteen (age 28.6 ± 3.2 years, height 183 ± 6.1 cm, body mass 80 ± 5.4 kg) professional Italian Serie A Soccer players (6 defenders, 6 midfielders, and 6 forwards) volunteered to participate in this study. Each player had at least 7 years of competitive experience in premiership. All the players were active members of the same Serie A team (ACF Fiorentina, Firenze, Italy) during the 2010–2011 season. Ten players were starters of their own country national A team.

The players trained 7 times a week throughout the preseason with a friendly match played on Thursday or during the weekend. Training sessions were mainly devoted to technical-tactical skill development with fitness training drills performed as single training sessions during the preseason. The training time during the observed period was 15 and 13% devoted to ball drills and generic aerobic-training, respectively. Twenty-one and 8% of the training time was spent training for technical-tactical skill development and matches, respectively. Mainly anaerobic training (strength

and sprint training, 5 and 9%, respectively) accounted for 14% of all the training time. Strength and power training was performed 2 times a week in a single morning session (3×8 – 10 repetition maximum weight-machine exercises). Strength training mainly aimed to prevent injury. The remaining time (29%) was spent doing warm-up routines.

Written informed consent was obtained from the players before the commencement of the study and after local Institutional Research Board research design approval.

During the 4 weeks preceding the study, the players abstained from heavy and supervised training (postchampionship break). Furthermore, to avoid potential confounding effects of prior exercise fatigue on the outcome variables, coaches were asked to ensure that their players refrained from heavy training the day preceding assessments. A record of the nutrient content was taken to provide sufficient carbohydrate intake during the week before the assessments. Throughout the study, all testing and training sessions took place at the same time of the day (between 9.00 and 13.00 hours) to avoid circadian influences.

Procedures

The players performed a 2-phase progressive treadmill test (Technogym Run Race 1400 HC, Gambettola, Italy) for the assessment of individual blood-lactate concentration profiles and maximal HR, respectively, on 2 occasions (at the start and after 8 weeks of training). The progressive treadmill test consisted of 4–5 submaximal exercise bouts at an initial running speed of $9 \text{ km} \cdot \text{h}^{-1}$ followed by a maximal incremental test to volitional fatigue. The treadmill running velocity was increased during the submaximal test by $1 \text{ km} \cdot \text{h}^{-1}$ every 5 minutes. Once capillary blood-lactate concentrations were elevated above $4 \text{ mmol} \cdot \text{L}^{-1}$, the treadmill speed was increased by $0.5 \text{ km} \cdot \text{h}^{-1}$ every 30 seconds until exhaustion (20,23). Capillary blood samples were taken from the earlobe immediately after each submaximal bouts and 3 minutes after exhaustion and analyzed to assess exercise blood-lactate concentrations using a portable amperometric micro-volume ($5 \mu\text{L}$) lactate analyzer (LactatePro, Arkray, Tokyo, Japan). Before each test, the analyzer was calibrated by following the manufacturer's recommendations.

The $\dot{V}\text{O}_{2\text{max}}$ was assessed using a progressive-maximal test completed on a 400-m athletic track, until exhaustion (26). Every 20 m, a cone was positioned as a reference. After an acoustic signal was activated, the subjects performed the incremental field test, starting from $8.0 \text{ km} \cdot \text{h}^{-1}$, with the speed then increasing by $0.5 \text{ km} \cdot \text{h}^{-1}$ every 140 m. The test ended when the participant failed twice to reach the next cone at the required time (objective evaluation) or he felt unable to cover another interval at the dictated speed (subjective evaluation). During the test, the players were verbally encouraged by the test leaders and coaches to provide maximal effort in the late stages of the test.

The $\dot{V}\text{O}_{2\text{max}}$ was considered to have been achieved on the attainment of at least two of the following criteria: (a)

a plateau in the $\dot{V}\text{O}_2$ despite increasing speeds, (b) a respiratory exchange ratio >1.10 , (c) an $\text{HR} \pm 10 \text{ b} \cdot \text{min}^{-1}$ of age-predicted HR_{max} ($208 - 0.7\text{age}$). Expired gases were analyzed using a breath-by-breath automated gas-analysis system (K4b², Cosmed, Rome, Italy) (10,11).

The highest HR measured in either maximal incremental test was used as the HR_{max} . Criteria for HR_{max} achievement were the attainment of subjective and visual exhaustion, blood-lactate concentrations $>8 \text{ mmol} \cdot \text{L}^{-1}$, and HR plateau attainment despite speed increments. The Yo-Yo IR1 performance was assessed according to the procedures reported by Castagna et al. (3).

Training volume and intensity were prescribed to players by team strength and conditioning and technical-tactical coaches. The prescribed training schedule represents the typical training program performed by professional premiership players to prepare for the upcoming competitive season (4).

During all the training sessions and tests, the HR was recorded every 5 seconds with a short-range telemetry system (Polar RS400 and Team-System, Polar Electro Oy, Kempele, Finland). The HR data were downloaded onto a portable PC and analyzed using dedicated software (Polar ProTrainer 5) and an electronic spreadsheet (Excel, Microsoft Corporation, USA).

Only aerobic-fitness and tactical skill development training sessions were monitored in this study. In all, 900 individual training sessions were monitored and used for calculations (average duration 90 ± 10 minutes). During all the training sessions, the HRs were recorded in each subject, and to avoid missing data were downloaded immediately at the end of each training session on a computer and analyzed using specific software (Polar ProTrainer 5) and an electronic spreadsheet (Microsoft Excel). To check for possible HR monitor malfunction (i.e., inability to record HR during exercise), the sport scientists supervising ($n = 3$) the data collection procedures checked HR displays (Polar RS400) during all the training sessions (4). Additionally, supervisors were available all the time to fix HR chest bands or to immediately substitute them. No missing data occurred during this study because of compliance to manufacturers' guidelines, proper device care, and passionate collaboration of all the players involved.

Statistical Analyses

The results are expressed as mean $\pm$ SD and 95% confidence intervals (95% CIs). Assumption of normality was verified using the Shapiro-Wilk W test. Variable association was assessed using Pearson's product-moment correlation coefficients. The qualitative magnitude of associations was reported as in Castagna et al. (4) as follows: trivial $r < 0.1$, small $0.1 < r < 0.3$, moderate $0.3 < r < 0.5$, large $0.5 < r < 0.7$, very large $0.7 < r < 0.9$, nearly perfect $r > 0.9$, and perfect $r = 1$.

Pretraining-to-posttraining changes were examined with paired t -tests. Cohen's d was used to assess the effect size

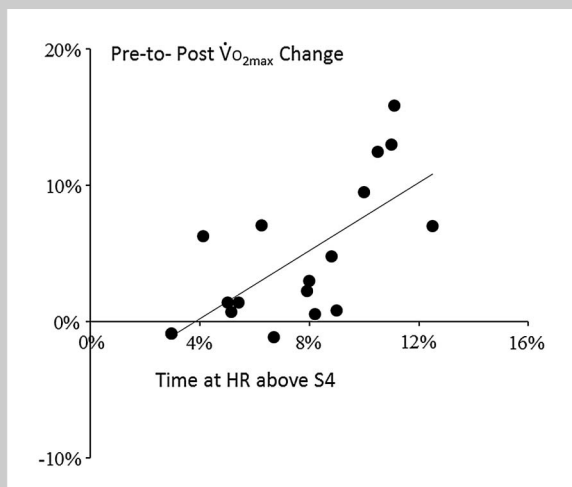


Figure 1. Relationship between time spent in the high-intensity zone (more than the heart rate at $4 \text{ mmol} \cdot \text{L}^{-1}$) and the pre to post percentage improvement in the $\dot{V}\text{O}_{2\text{max}}$ ($r = 0.65$, $p = 0.02$; 95% confidence interval 0.16–0.89, $n = 18$).

(ES) (6). Effect sizes of >0.8 , between 0.8 and 0.5, between 0.5 and 0.2, and <0.2 were considered as large, moderate, small, and trivial, respectively. Significance was set at $p \leq 0.05$.

The intraclass correlation coefficient for the fitness measurements ranged between 0.89 and 0.92 in a preliminary quality control performed before the commencement of the study with a population similar to this study (i.e., semi-professional soccer players, $n = 16$).

RESULTS

Analysis of a total of 900 individual training sessions showed that soccer players exercised for 73.6 ± 3.7 (2,945 $\pm$ 148 minutes), 19.1 ± 3.5 (763 $\pm$ 141 minutes),

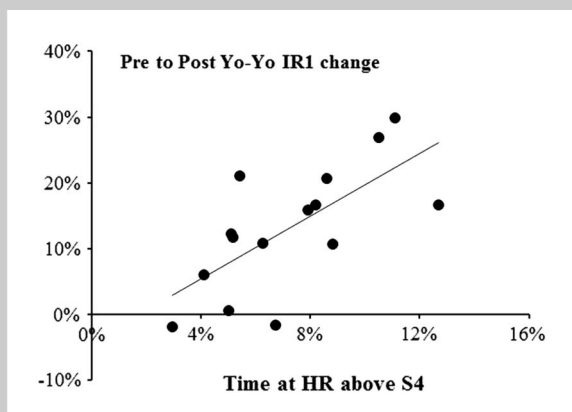


Figure 2. Relationship between time spent in the high-intensity zone (more than the heart rate at $4 \text{ mmol} \cdot \text{L}^{-1}$) and the pre to post percentage improvement in the Yo-Yo IR1 ($r = 0.69$, $p = 0.01$; 95% confidence interval 0.17–0.89, $n = 18$).

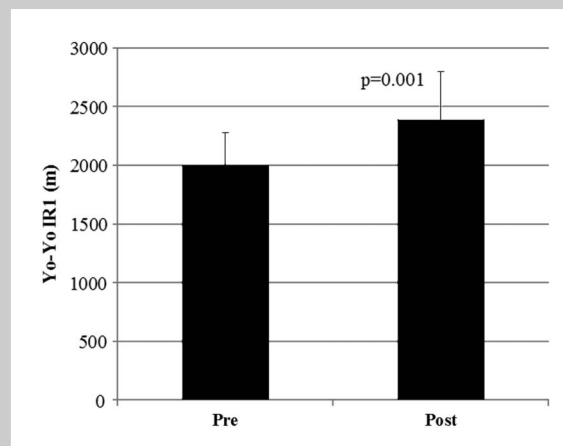


Figure 3. Pretraining-to-posttraining changes in the Yo-Yo IR1 performance ($n = 18$).

and $7.3 \pm 2.9\%$ (292 $\pm$ 116 minutes) of the total training time at low, medium, and high intensities, respectively ($p < 0.001$). Heart rates at 2 and $4 \text{ mmol} \cdot \text{L}^{-1}$ were 82.0 ± 3.1 and $91.9 \pm 3.0\%$ of the HRmax, respectively.

The S2 (from 10.9 ± 1.2 to $12.0 \pm 1.1 \text{ km} \cdot \text{h}^{-1}$; $p = 0.0001$; 95% CI from 0.73 to $1.60 \text{ km} \cdot \text{h}^{-1}$; ES = 1.09), S4 (from 13.9 ± 2.0 to $14.9 \pm 1.5 \text{ km} \cdot \text{h}^{-1}$; $p = 0.0008$; 95% CI from 0.48 to $1.52 \text{ km} \cdot \text{h}^{-1}$; ES = 1.09), and $\dot{V}\text{O}_{2\text{max}}$ (from 58.2 ± 4.4 to $61.4 \pm 4.1 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$; $p = 0.009$; 95% CI from 0.75 to $4.3 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$; ES = 0.9) significantly improved pretraining to posttraining. The Yo-Yo IR1 performance significantly ($p = 0.001$) improved after the training period (from $2,000 \pm 279$ to $2,390 \pm 409 \text{ m}$, 95% CI from 237 to 498 m; ES = 2.1).

The training time spent at high intensity showed a significantly large ($r = 0.65$, $p = 0.02$, 95% CI 0.16–0.89)

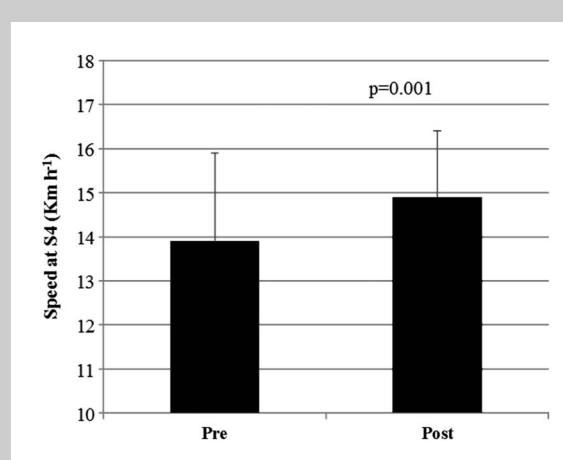


Figure 4. Pretraining-to-posttraining changes in speed at $4 \text{ mmol} \cdot \text{L}^{-1}$ performance ($n = 18$).

relationship with $\dot{V}O_{2\max}$ improvements (Figure 1). Changes in the speed at S2 and S4 showed a significant very large and large association with the time spent at high intensity during training ($r = 0.78$, $p = 0.002$, 95% CI 0.39–0.93; $r = 0.60$, $p = 0.03$, 95% CI 0.07–0.86). The Yo-Yo IR1 performance improvements were largely associated with the time spent at high intensity ($r = 0.66$, $p = 0.01$, 95% CI 0.17–0.89; Figure 2).

No significant relationships were observed between time spent at low- and medium-intensity HR zones during training and any of the aerobic fitness and performance variables considered in this study.

DISCUSSION

The main finding of this study was the significant and practical effect of training time spent at intensities $>90\%$ of the individual HRmax on aerobic fitness and performance variables in professional well-trained soccer players.

These findings are in line with those of previous randomized experimental training studies that over a similar observational period imposed generic or specific training at the HR in the range of 90–95% of the HRmax in young non-professional soccer players (20,31).

Nonexperimental and randomized training studies have been recently proposed as a viable method to evaluate training responses in elite-standard professional soccer over the preseason (4). Despite the practical interest for this kind of methodological approach, the associated build-in limitation is a resulting lower external validity (32). As a result, study replications may act as a viable way to test the variables of interest for external validity (i.e., inductive approach) (32).

This study results are similar to those previously reported by Castagna et al. (4) in a team study performed over a similar observational period (i.e., 6 vs. 8 weeks) with Italian professional elite-standard (Italian Serie A) male soccer players during the preseason. Indeed, in that study, improvements in submaximal aerobic fitness (i.e., speeds at 2 and 4 $\text{mmol} \cdot \text{L}^{-1}$) were reported to be associated with the time spent at HR at or $>90\%$ of individual HRmax. Furthermore, the HR distribution reported in the Castagna et al. (4) training study was remarkably similar to that reported in this article. Interestingly, either studies showed that improvement in aerobic fitness can be achieved totaling at least 7–8% of the entire training time at or $>90\%$ of HRmax. These findings' replication provide evidence as per TL polarization in professional men's soccer (4).

The effectiveness (i.e., achievement of supposed improvements) of a training program is warranted by a dynamic balance between intensity and volume (8,9).

Given this, the difference in training volume between the nonexperimental training studies should not be neglected to understand possible “dose-response” guidelines. Although there are several methods to express training volume session duration has been suggested to be a valid approach (4,14,15).

In the Castagna et al. (4) study, the performed training volume (i.e., total training time) in the selected HR zones was 52–57% of this study time. This was partially so because of the 33% longer observation time considered in this study (i.e., 6 vs. 8 weeks) and associated higher number of training sessions monitored and included in the calculations (i.e., 504 vs. 900 monitored training sessions, +78%) (4). Over 4 weeks of training, Impellizzeri et al. (20) reported significant improvement of aerobic fitness having players exercising at 90–95% of the HRmax for 6% of the total training time (i.e., 2,491 minutes). This was a total training volume quite lower than that reported in this study but greater than that reported in the Castagna et al. (4) study (i.e., +19%). These findings suggest that different training volume magnitudes may lead to similar improvements in aerobic fitness. This supports the interest of the time devoted to high-intensity training in soccer (4,20,21,31).

The available body of evidence suggests that time at 90–95% of individual HRmax should be in the range of 6–8% of the total training time to be effective for aerobic-fitness development (4,20,21,31).

However, the magnitude of the improvements does not seem to be related to the absolute time spent in the high-intensity zone as per the results reported in the cited studies. In this regard, further investigations examining the effect of volume variation in time spent in the aerobic-fitness beneficial HR zones are warranted. This would be of great interest to more efficiently prescribe effective training in soccer (4).

Again similarly to what was previously reported in soccer randomized controlled training studies, the results of this research showed that the time spent in high-intensity HR zones produced a significant improvement in the whole range of the variables considered as the main component of aerobic fitness (4,20,21,31). However, the nonsystematic training intervention considered in this study showed a slightly lower improvement in the $\dot{V}O_{2\max}$ and submaximal aerobic fitness compared with experimentally implemented generic or specific training interventions in men's soccer (4,20,21,31). Difference in baseline fitness may partially account for the reported difference in aerobic-fitness improvements. However, it could be speculated that the different training volumes considered in the various studies may have played a role as well. This induces a transitory state of functional overreaching in the players that may have in some way hidden the actual training response (i.e., test results) as a consequence of the provided TLs (8,9).

A remarkable match-to-match variability in game activities has been reported in elite-standard soccer (16). This suggests the need for power calculations to gain meaningful information as per activity category reliability. A practical way to limit the reported variation in match-to-match activities is the use of experimentally organized friendly matches or consideration for criterion-performance tests that may enable the track of fitness variations to estimate players'

capability to cope with game demands (i.e., permanence at high intensity). In this study, as match criterion performance was considered the performance (i.e., distance covered) in the Yo-Yo IR1 (1).

The improvement in the Yo-Yo IR1 found in this study was lower (i.e., 12.5 vs. 31%) than those reported in training studies considering soccer officials (i.e., +31%) and young soccer players (i.e., +17–22% improvements) (2,18,22). The reported difference in Yo-Yo IR1 improvement may be because of the baseline fitness level reported in the players of this study that was approximately similar to those achieved by the participants of the cited studies at the end of the considered training interventions (2,18,22,29). Similarly to aerobic fitness, the training time spent with HR >90% of the HRmax positively affected Yo-Yo IR1 performance variation (i.e., HR responsiveness, Figures 1–2) (19).

In light of the findings of this study, high-intensity training may be considered as a way to positively affect aerobic fitness and players' capability to cope with game demands in men's elite-standard professional soccer players.

The main limitation of this study was the use of a convenience sample (i.e., a professional soccer team), and consequently, this investigation should be regarded as a case study (4). However, the provided confirmation of results reported in previously published team studies conducted over the same seasonal training period may suggest the validity of the used observational design. Conclusive inferences about the precompetitive season training intensity profile used by elite professional soccer players should be drawn studying larger randomized cohorts of players or with this study's replications to warrant inductive evidence.

PRACTICAL APPLICATIONS

This study results provided further evidence on the effectiveness of intensity rather than volume training in developing aerobic fitness and performance in male soccer players. In this regard, weekly doses of 6–8% of total training time devoted at 90–95% of the HRmax seem to be effective in aerobic fitness and performance development. Anyway, differences in absolute training (i.e., minutes spent in each intensity zone) time should be considered when addressing this issue.

Given the diversity of training responses that may be expected in soccer because of group-specific vs. generic training, the importance of monitoring training intensity is warranted (20). In this regard, this study provided further supporting evidence for the use of HR (i.e., HR responsiveness) to assess the individual responses to training in soccer players (4,12,20,21). Consequently, HR monitoring can be considered as an appropriate tool to quantify and modulate the individual TL in soccer players at the professional level.

This study and Castagna et al. (4) showed that the descriptive longitudinal designs were a viable approach for studying the preseason development of aerobic fitness and performance in elite-standard male soccer players using

a minimally invasive method. As a result, further studies addressing different aspects of fitness and performance in professional soccer with this approach are warranted.

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